

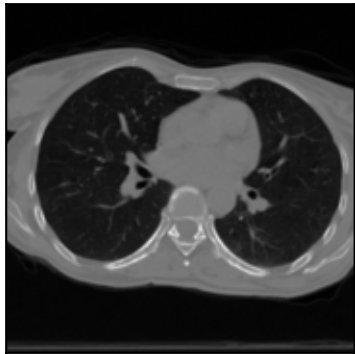
DVF QA pipeline report

DIR-Lab-case1 [frame_diff]

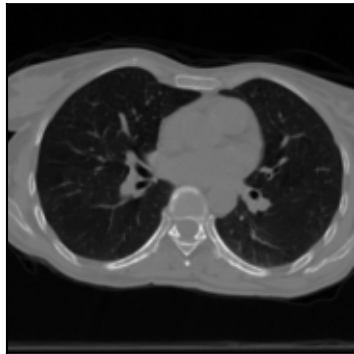
Field	Value
Case	DIR-Lab-case1 [frame_diff]
Generated	2026-05-20 08:01:37
source	Case1Pack
lateral_strategy	frame_diff
fps	15.0
n_cycles	4
frames_per_cycle	30
n_phases	10
ct_shape	(94, 256, 256)
spacing_xyz_mm	(0.97, 0.97, 2.5)
detector_shape	(256, 256)
pixel_spacing_mm	(1.0, 1.0)

4DCT phase preview

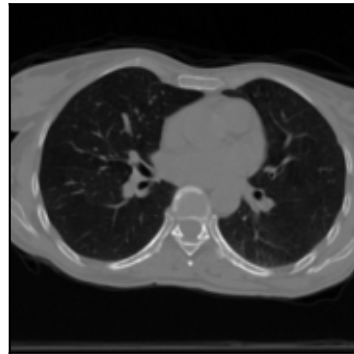
φ_0



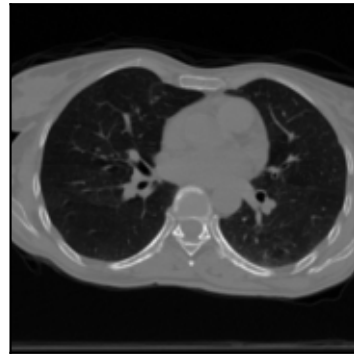
φ_1



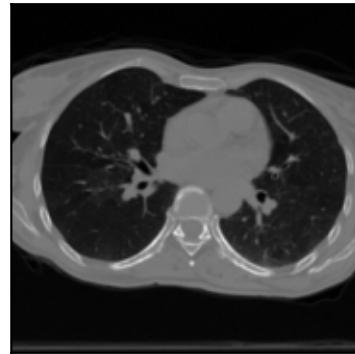
φ_2



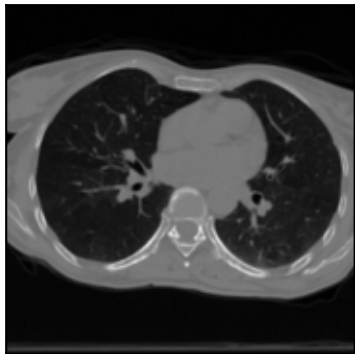
φ_3



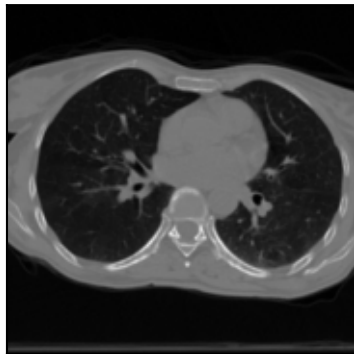
φ_4



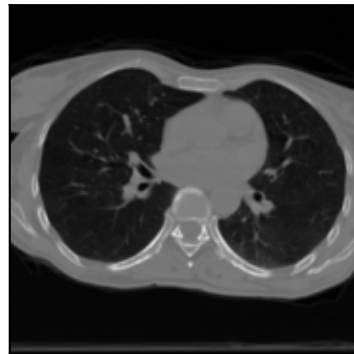
φ_5



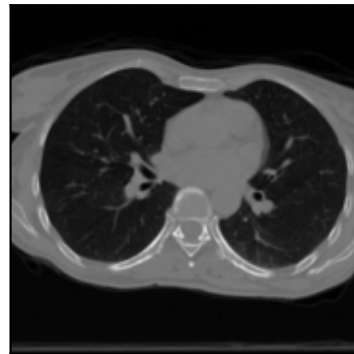
φ_6



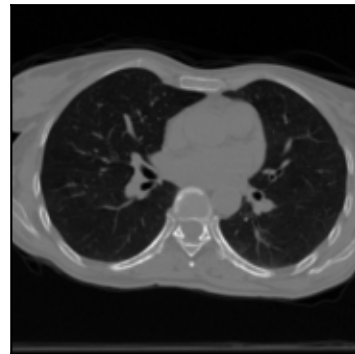
φ_7



φ_8



φ_9



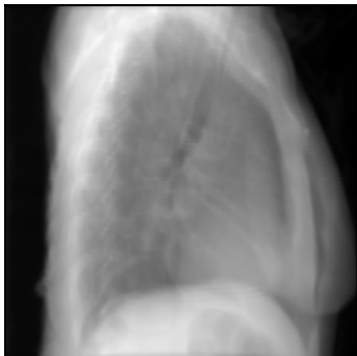
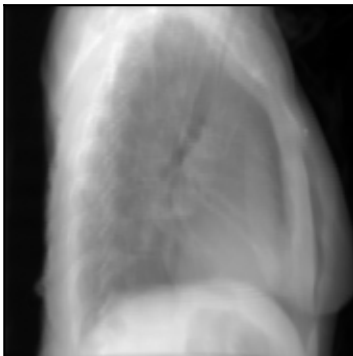
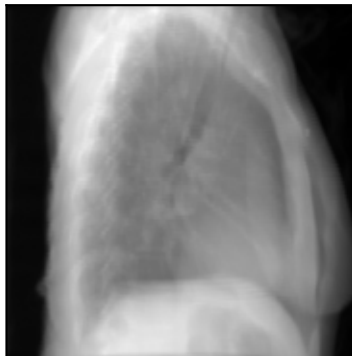
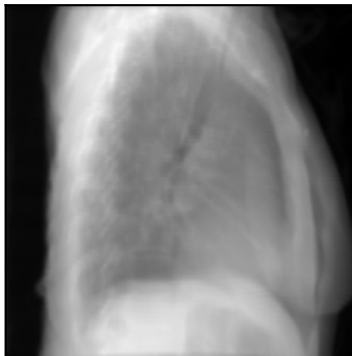
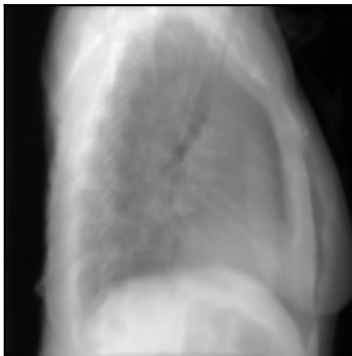
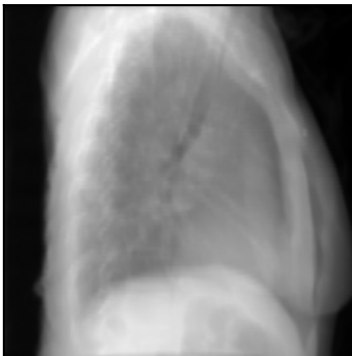
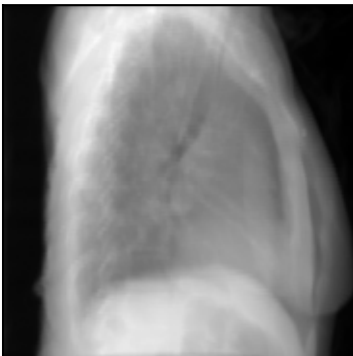
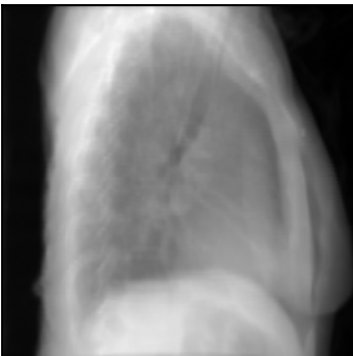
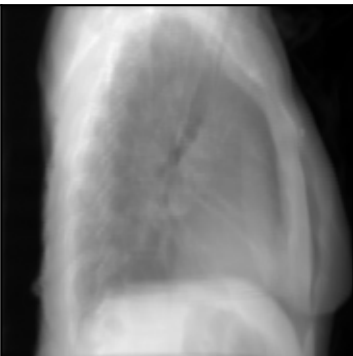
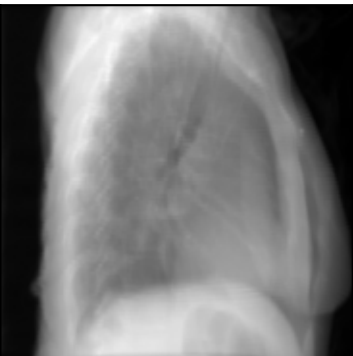
Per-phase DRRs

AP φ_0 AP $\phi 1$ AP ϕ_2 AP $\phi 3$ AP ϕ_4 

AP $\phi 5$



AP φ6

AP $\phi 7$ AP $\varphi 8$ AP ϕ_9 Lat ϕ_0 Lat ϕ_1 Lat ϕ_2 Lat ϕ_3 Lat ϕ_4 Lat ϕ_5 Lat ϕ_6 Lat ϕ_7 Lat ϕ_8 Lat ϕ_9 

Dynamic sequence samples

AP t=0 (0.00s)



AP t=24 (1.60s)



AP t=48 (3.20s)



AP t=72 (4.80s)



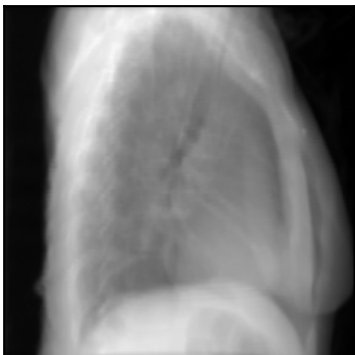
AP t=96 (6.40s)



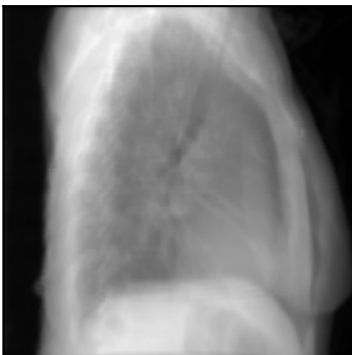
AP t=120 (8.00s)



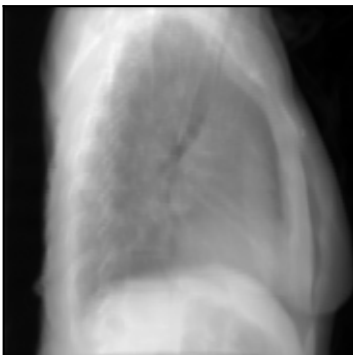
Lat t=0



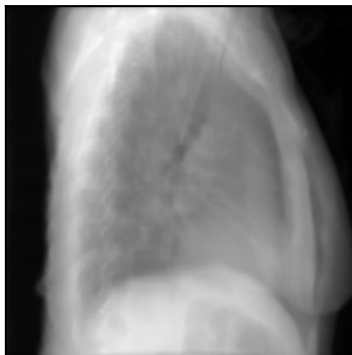
Lat t=24



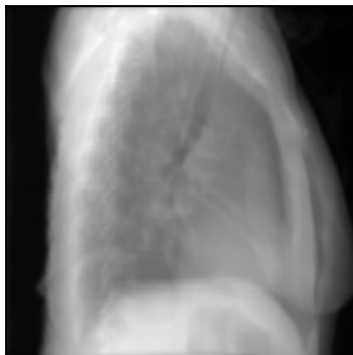
Lat t=48



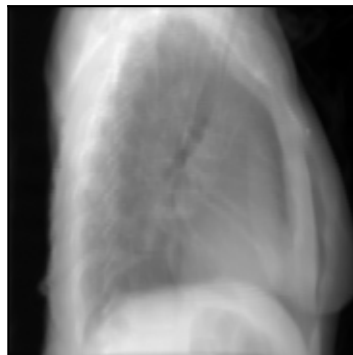
Lat t=72



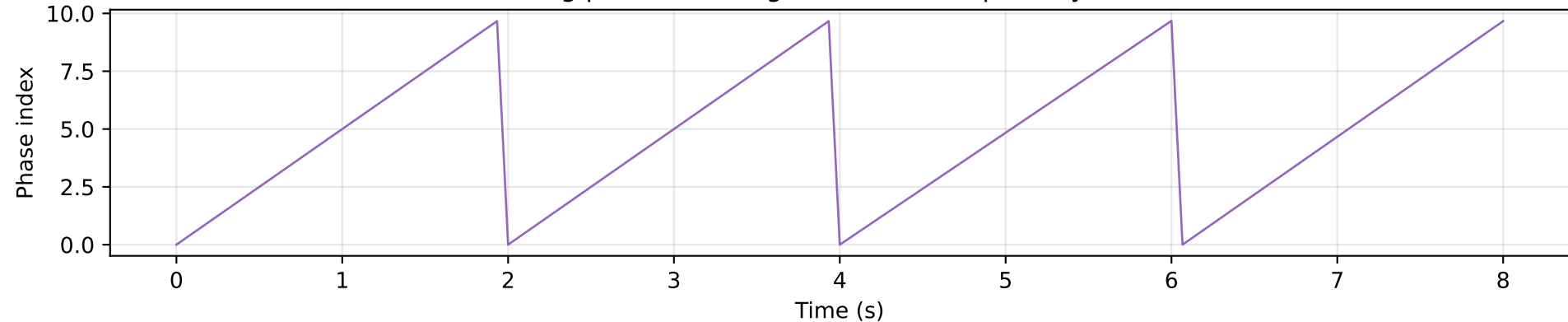
Lat t=96



Lat t=120

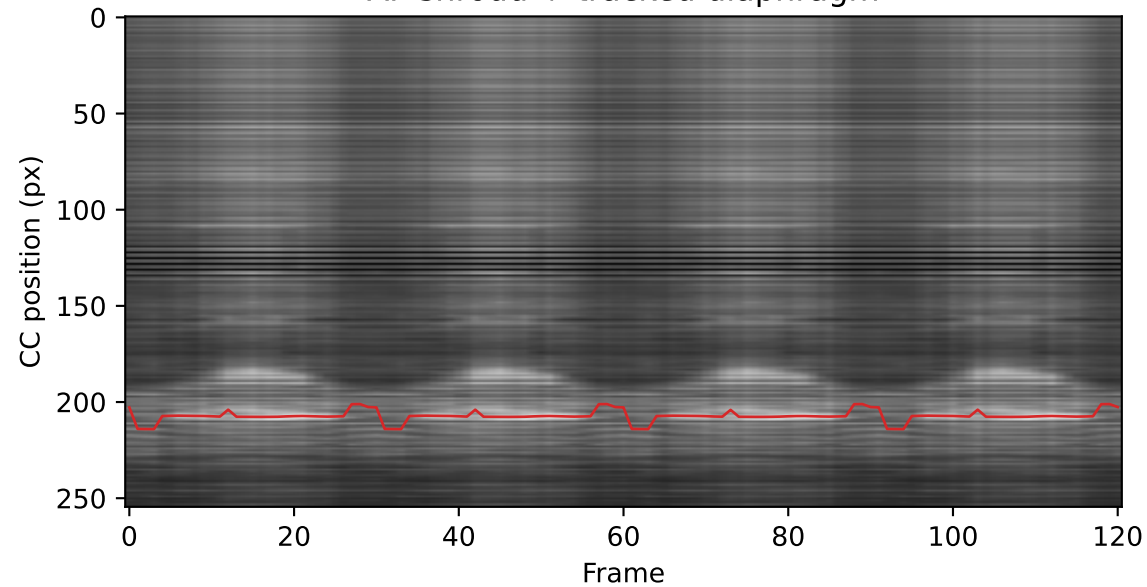


Driving phase curve
Driving phase curve (ground-truth respiratory motion)

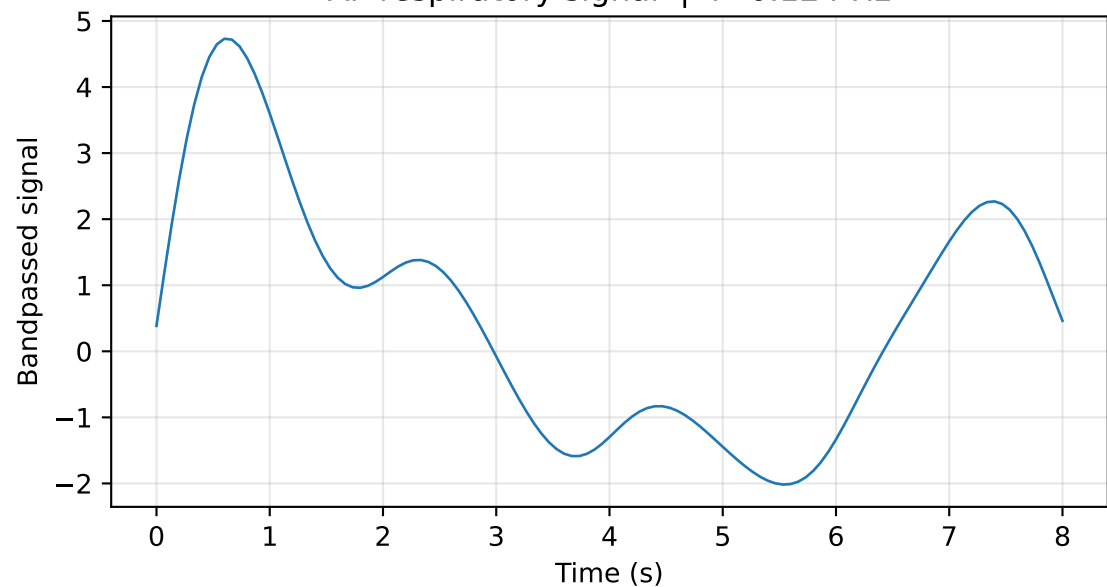


Amsterdam Shroud — AP

AP shroud + tracked diaphragm

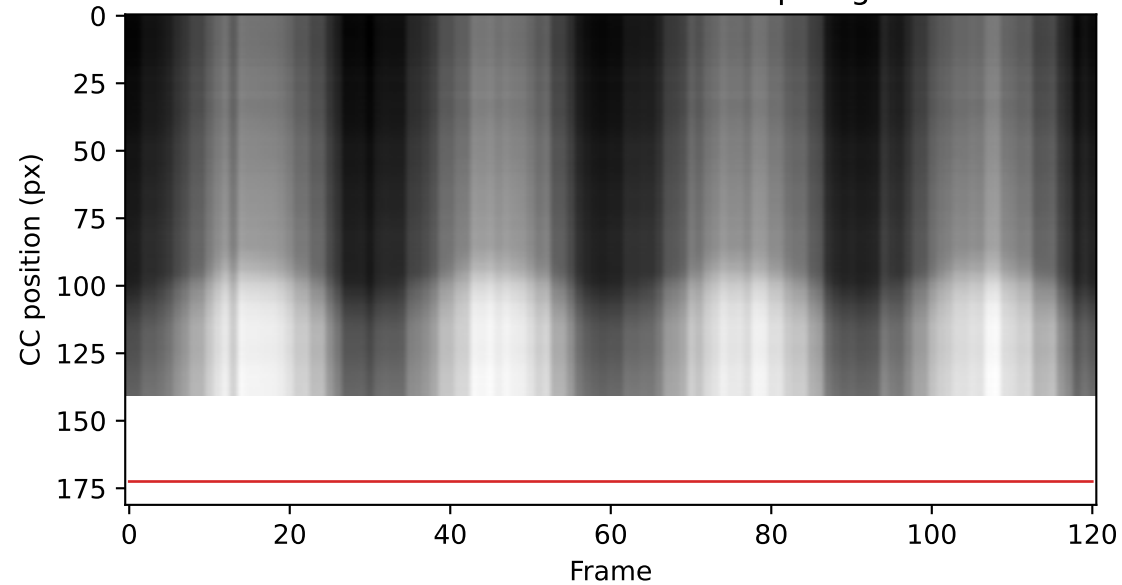


AP respiratory signal | $f=0.124$ Hz

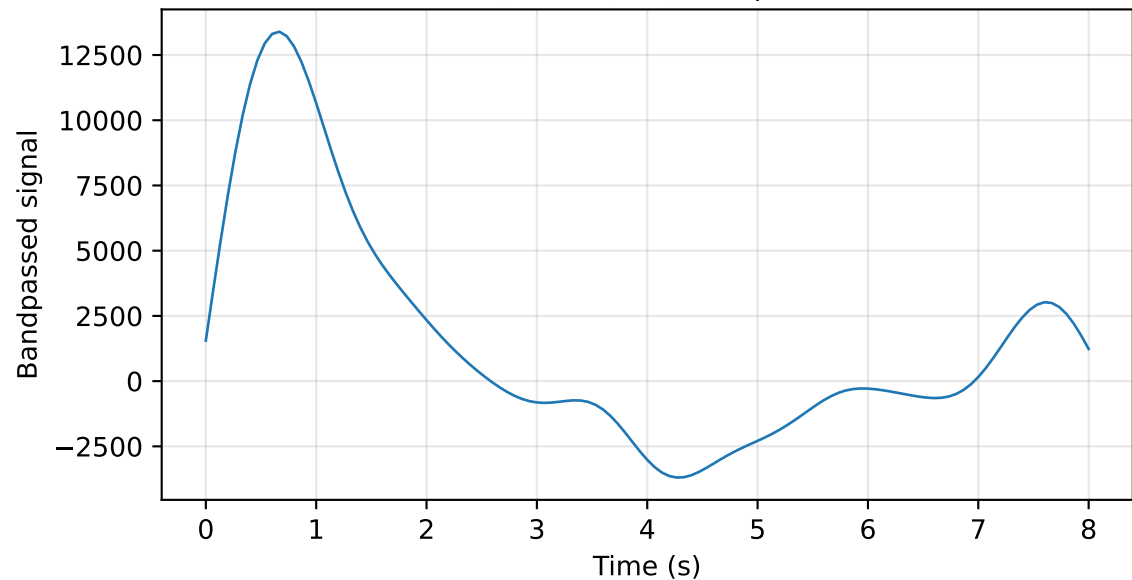


Amsterdam Shroud — Lateral

Lateral shroud + tracked diaphragm

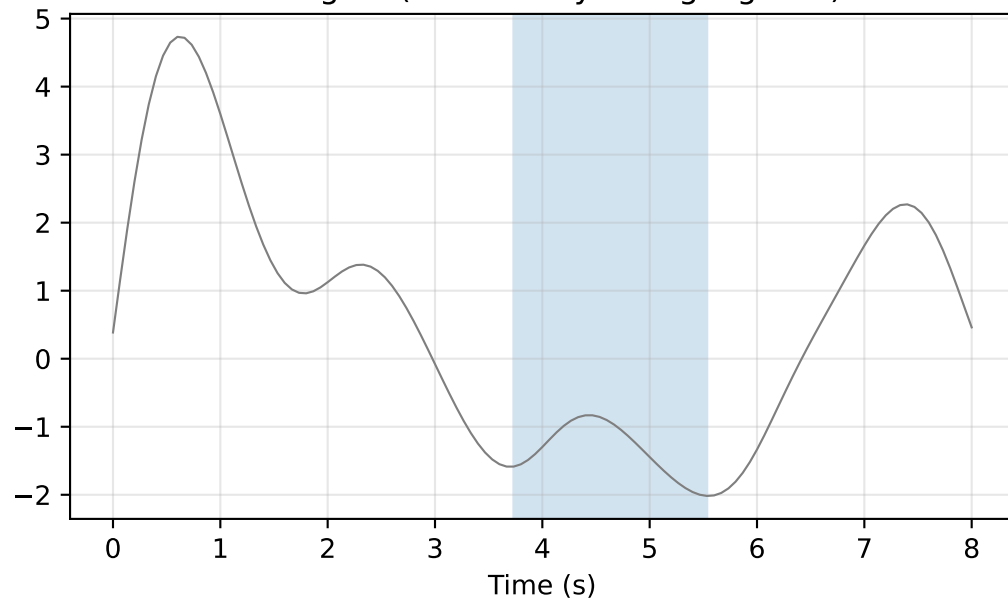


Lateral respiratory signal | $f=0.124$ Hz

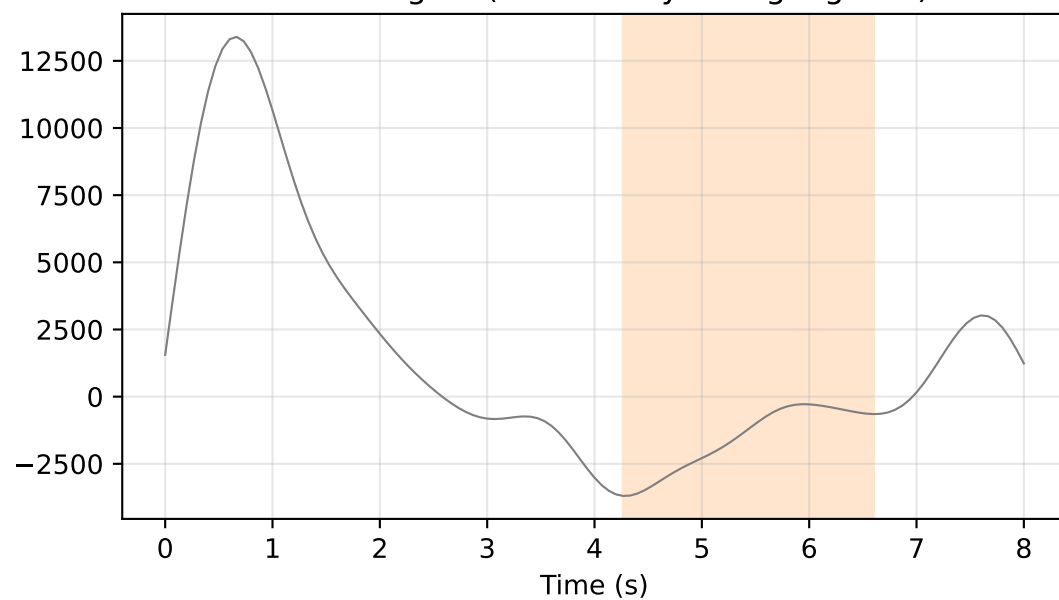


Cycle pairing

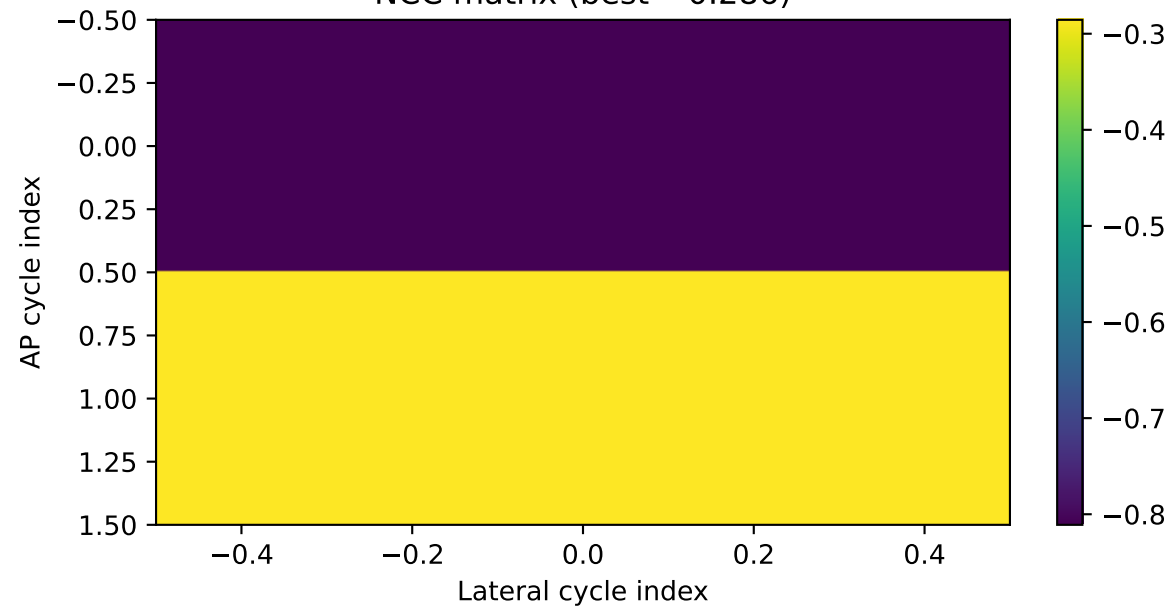
AP signal (selected cycle highlighted)



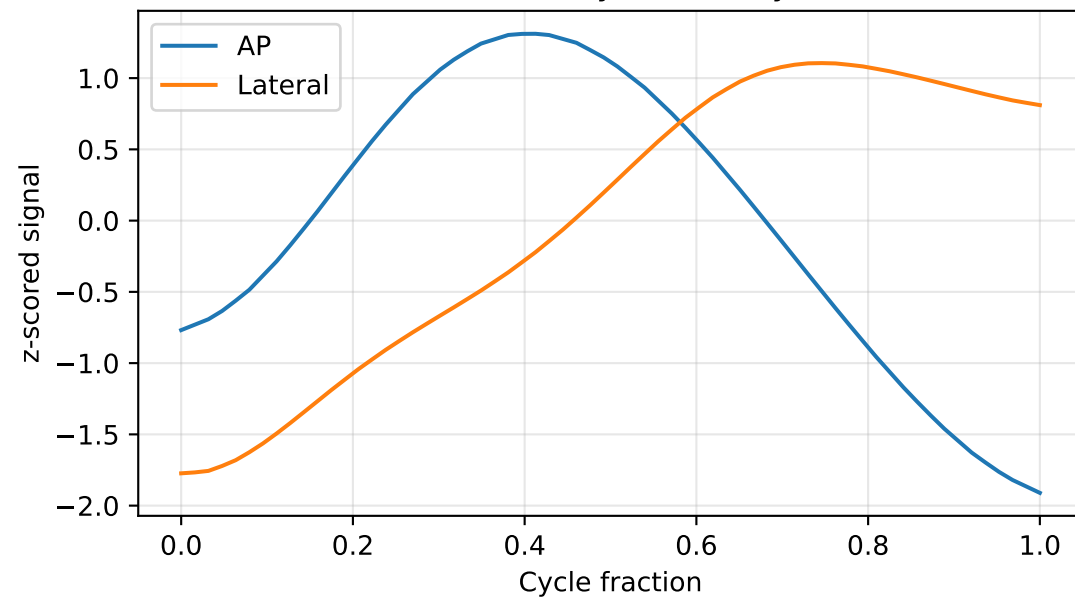
Lateral signal (selected cycle highlighted)



NCC matrix (best=-0.286)



Selected cycle overlay



Metrics

Metric	Value
AP dominant frequency (Hz)	0.1240
AP period (frames)	121.00
Lateral dominant frequency (Hz)	0.1240
Lateral period (frames)	121.00
Best pair NCC	-0.2855
AP cycle frames	56 - 83
Lateral cycle frames	64 - 99